Mini-Activity-4-Solutions.R

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```
##### Mini-Activity 4 Solutions #####
```

We begin by setting up our environment for the assignment.

```
# Load libraries
library(rstatix)
```

```
##
## Attaching package: 'rstatix'

## The following object is masked from 'package:stats':
##
##     filter
```

```
library(psych)
library(ez)
```

```
## Registered S3 method overwritten by 'lme4':
##   method           from
##   na.action.merMod car
```

```
library(here)
```

```
## here() starts at /Users/gcrone14/Desktop/R Projects/PSYC3031-F26
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x ggplot2::%+%()   masks psych::%+%()
## x ggplot2::alpha() masks psych::alpha()
## x dplyr::filter()  masks rstatix::filter(), stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
# Import data
worry <- read_csv(here::here("Data/worry.csv"))
```

```
## Rows: 30 Columns: 5
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (5): ID, time1, time2, time3, time4
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
glimpse(worry)
```

```
## Rows: 30
## Columns: 5
## $ ID    <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 1~
## $ time1 <dbl> 65, 69, 60, 57, 62, 67, 63, 78, 70, 71, 81, 57, 68, 64, 71, 73, ~
## $ time2 <dbl> 66, 66, 64, 54, 62, 66, 58, 82, 68, 65, 85, 63, 57, 72, 72, 65, ~
## $ time3 <dbl> 63, 62, 59, 59, 62, 60, 60, 83, 66, 62, 83, 59, 55, 68, 66, 62, ~
## $ time4 <dbl> 63, 65, 59, 58, 54, 57, 53, 80, 65, 65, 86, 61, 53, 67, 65, 59, ~
```

```r
# Clean data (convert from wide to long,
# convert IDs to factor)
worry_final <- worry |>
  pivot_longer(cols = time1:time4,
               names_to = "time",
               values_to = "DEW_score") |>
  mutate(ID = factor(ID),
         time = factor(time,
                       levels = paste0("time", 1:4),
                       labels = c("Just married", "5 years",
                                  "10 years", "15 years"))
  )
glimpse(worry_final)
```

```
## Rows: 120
## Columns: 3
## $ ID        <fct> 1, 1, 1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, ~
## $ time      <fct> Just married, 5 years, 10 years, 15 years, Just married, 5 y~
## $ DEW_score <dbl> 65, 66, 63, 63, 69, 66, 62, 65, 60, 64, 59, 59, 57, 54, 59, ~
```

1)

Do men tend to express worries more (or less) to their wives the longer they are married?

2)

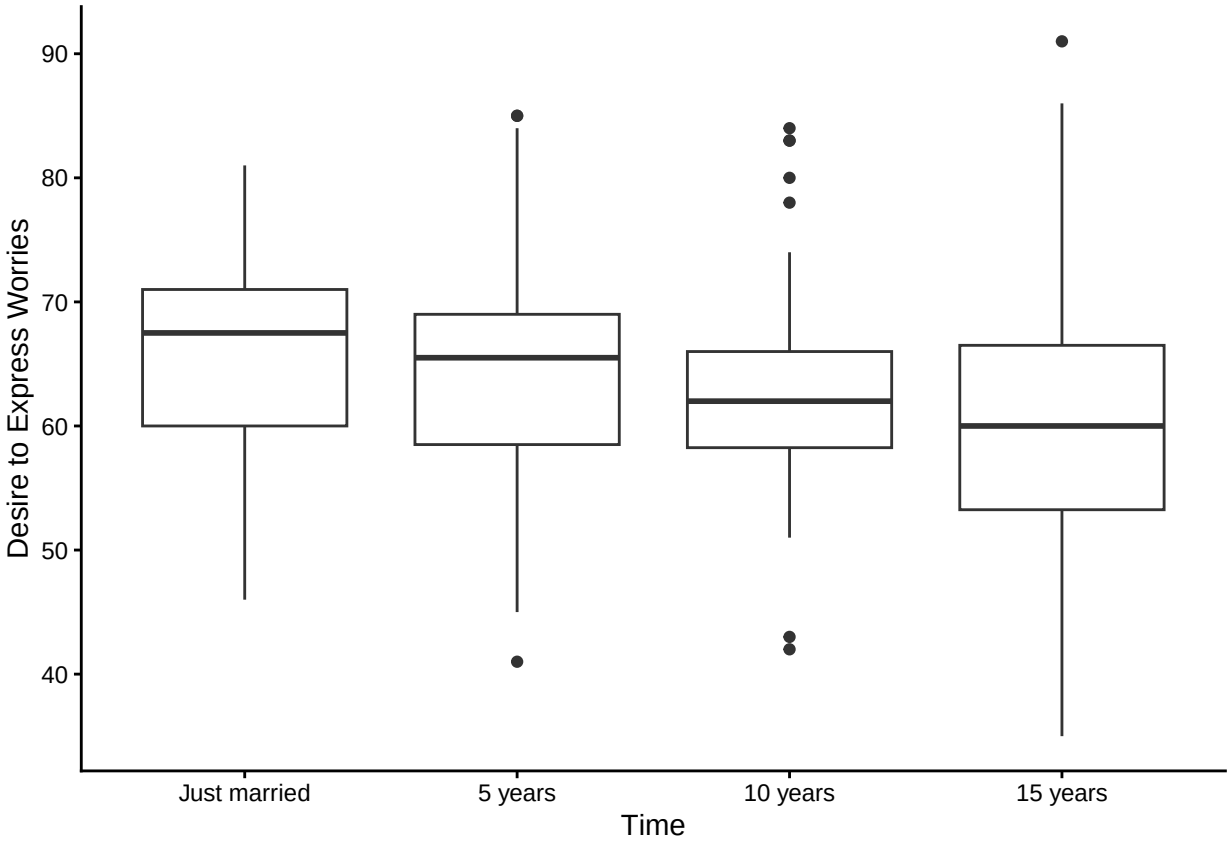
Let's explore the differences as a function of time.

```
worry_final |>
  select(DEW_score) |>
  describeBy(worry_final$time)
```

```
##
##  Descriptive statistics by group
## group: Just married
##           vars  n mean   sd median trimmed  mad min max range skew kurtosis
## DEW_score    1 30 65.8 9.23   67.5   66.12 8.15  46  81    35 -0.3    -0.78
##             se
## DEW_score 1.68
## ------------------------------------------------------------
## group: 5 years
##           vars  n  mean    sd median trimmed mad min max range skew kurtosis
## DEW_score    1 30 65.43 10.69   65.5   65.38 8.9  41  85    44    0    -0.18
##             se
## DEW_score 1.95
## ------------------------------------------------------------
## group: 10 years
##           vars  n mean    sd median trimmed  mad min max range skew kurtosis
## DEW_score    1 30 63.1 10.68     62   62.79 5.93  42  84    42 0.33    -0.28
##             se
## DEW_score 1.95
## ------------------------------------------------------------
## group: 15 years
##           vars  n  mean    sd median trimmed   mad min max range skew kurtosis
## DEW_score    1 30 61.93 12.57     60   61.25 10.38  35  91    56 0.41    -0.21
##             se
## DEW_score 2.29
```

There seems to be a decrease in men's desire to express their worries to their wives as the marriage goes on. Let's plot this.

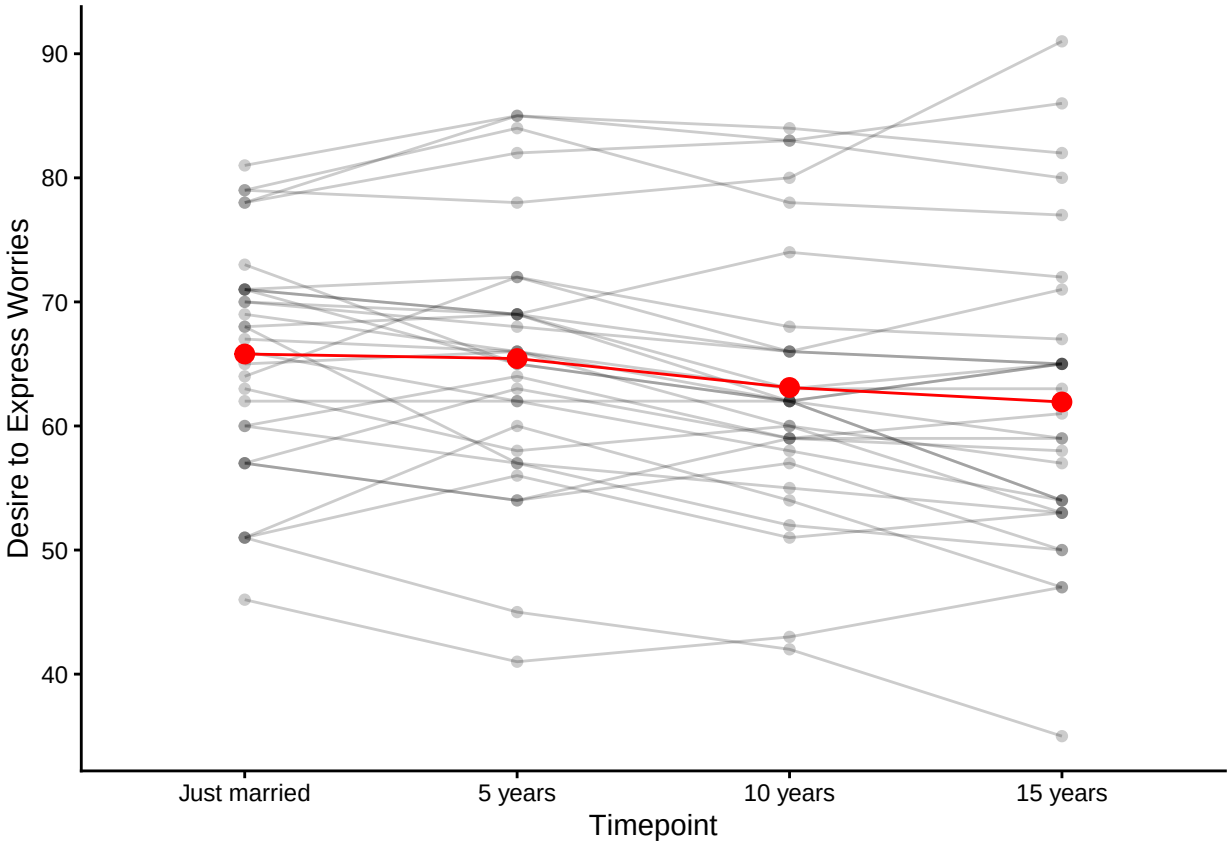
```
worry_final |>
  ggplot(aes(x = time, y = DEW_score)) +
  geom_boxplot() +
  labs(x = "Time",
       y = "Desire to Express Worries") +
  theme_classic()
```



There does appear to be a slight decrease over time, but it's not particularly strong, and there seems to be more variability in responses as the marriage goes on.

Perhaps a spaghetti plot will further elucidate this?

```
worry_final |>
  ggplot(aes(x = time, y = DEW_score)) +
  geom_point(alpha = 0.2) +
  geom_line(aes(group = ID), alpha = 0.2) +
  stat_summary(geom = "point", fun = mean,
               color = "red", size = 3) +
  stat_summary(aes(group = 1),
               geom = "line", fun = mean,
               color = "red") +
  labs(x = "Timepoint", y = "Desire to Express Worries") +
  theme_classic()
```



The spaghetti plot affirms this small decrease on average over time, with some men showing an increased desire to express worries, and some showing the opposite.

In APA:

As the length of marriages went on, men tended to, on average, express fewer worries to their wives as measured by the DEW scale (Just Married: $M = 65.9$, $SD = 9.23$; 5 years: $M = 65.4$, $SD = 10.69$; 10 years: $M = 63.1$, $SD = 10.68$; 15 years: $M = 61.93$, $SD = 12.57$).

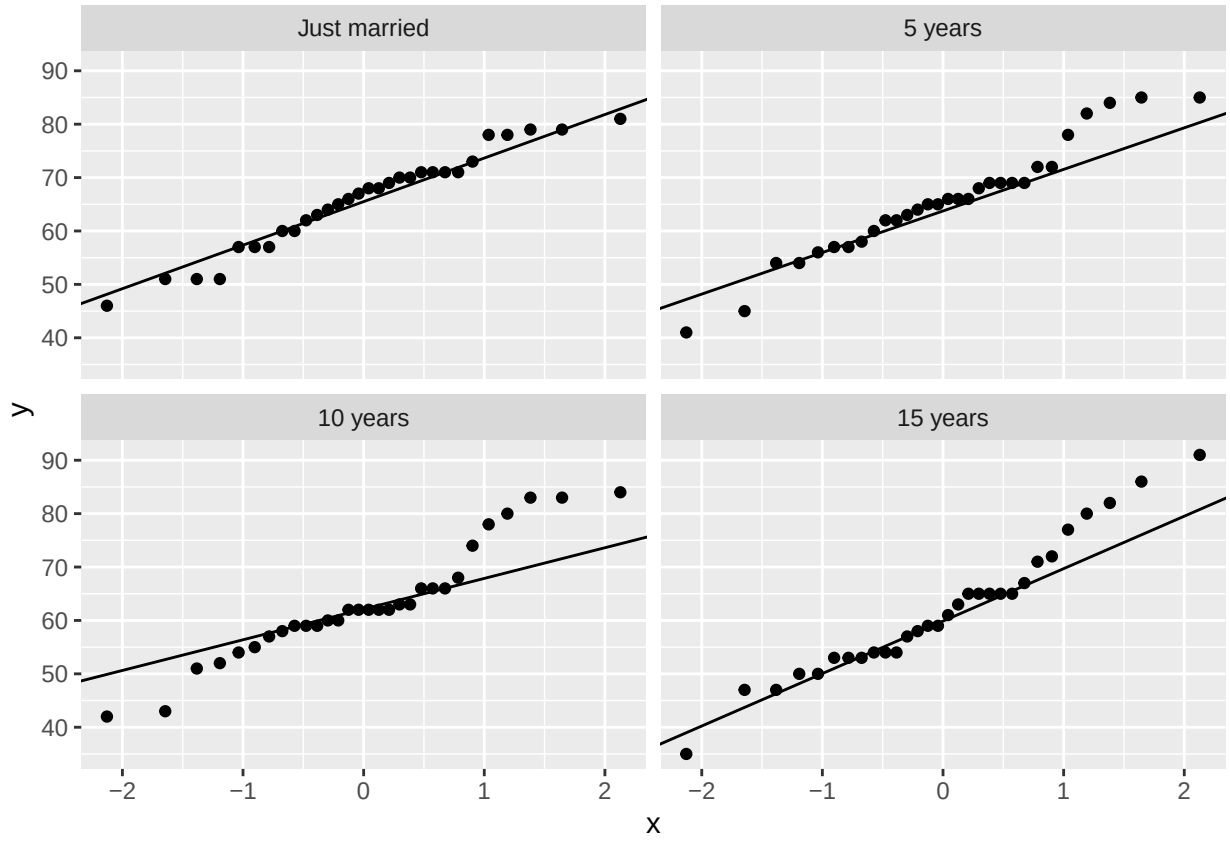
3)

We assess normality and sphericity.

Note that the assumption of independence is inherently violated, because the data are nested (people are nested within repeated-measures), but we can assume that individuals in the study were randomly selected to take part.

Normality can be assessed with a Q-Q plot.

```
worry_final |>
  ggplot(aes(sample = DEW_score)) +
  geom_qq() +
  geom_qq_line() +
  facet_wrap(~time)
```



The data align with the Q-Q plots well, so the assumption of normality seems met. (I wonder if the data were simulated...)

Sphericity can be assessed using McCauchly's test for sphericity.

```
res <- ezANOVA(data = worry_final,
               wid = ID,
               # New argument: within
               # specifies the within-subjects
               # variable, indicating that the
               # observations are DEPENDENT
               within = time,
               dv = DEW_score)

res$`Mauchly's Test for Sphericity`
```

```
##   Effect         W           p p<.05
## 2   time 0.4831003 0.001171425     *
```

The assumption of sphericity is, however, violated, since the Mauchly's test is statistically significant.

Thus, we'll need to use the sphericity-adjusted estimates when interpreting the eventual ANOVA.

```
res$`Sphericity Corrections`
```

```
##   Effect       GGe        p[GG] p[GG]<.05       HFe        p[HF] p[HF]<.05
## 2   time 0.6979155 0.0009104655         * 0.7533862 0.0006378608         *
```

Given that both epsilon estimates are in different ranges, we can't use our recommendation blindly.

Whichever correction you use is fine, so long as you justify it.

I'll use the Greenhouse-Gaiser correction because it's more conservative (so less likely to make a Type I error). The epsilon value is 0.70, which is statistically significant, $p < .0001$.

Thus, the average DEW scores differ across at least two time points.

4) We already interpreted the epsilon values and their p-values to account for the sphericity violation. Since the ANOVA was technically statistically significant, we can conduct pairwise tests.

```
pairwise_res <- pairwise_t_test(data = worry_final,
                                formula = DEW_score ~ time,
                                paired = TRUE,
                                p.adjust.method = "bonferroni",
                                detailed = TRUE) |>
  select(-p) |>
  # organize output from highest-to-lowest differences
  arrange(desc(estimate))
pairwise_res |>
  select(estimate, group1, group2, p.adj,
         p.adj.signif)
```

```
## # A tibble: 6 x 5
##   estimate group1       group2     p.adj p.adj.signif
##      <dbl> <chr>        <chr>      <dbl> <chr>
## 1    3.87  Just married 15 years 0.0328  *
## 2    3.5   5 years      15 years 0.0119  *
## 3    2.7   Just married 10 years 0.0423  *
## 4    2.33  5 years      10 years 0.00456 **
## 5    1.17  10 years     15 years 0.914   ns
## 6    0.367 Just married 5 years  1       ns
```

```
# Double-check range of DEW_score
range(worry_final$DEW_score)
```

```
## [1] 35 91
```

All mean differences were statistically significant, except for the difference between 10 and 15 years, and the difference between just married and 5 years.

However, it is crucial to note that these mean differences are *very small*: the largest difference is only 3.87, which is tiny given that the DEW scale ranges from 35 to 91.

So even though several mean differences are statistically significant, it's not very meaningful in practice. Practically, a ~4-unit difference in DEW may mean almost nothing.

APA results:

Janice was interesting in determining if the average man tended to express their worries more (or less) the longer they are married, so I (kindly) ran a repeated-measures ANOVA (for her).

Descriptively, as the length of marriages went on, men tended to, on average, express fewer worries to their wives as measured by the DEW scale (Just Married: $M = 65.9$, $SD = 9.23$; 5 years: $M = 65.4$, $SD = 10.69$; 10 years: $M = 63.1$, $SD = 10.68$; 15 years: $M = 61.93$, $SD = 12.57$).

Prior to running the repeated-measures ANOVA, we checked both the assumptions of normality and sphericity, assuming that the assumption of random sampling was met. While the assumption of normality was met (given high alignment of observed data to the Q-Q line for data on the outcome variable at each time point), the assumption of sphericity was not, Mauchly's Test: W = 0.48, p = .001.

Thus, instead of using the traditional ANOVA F-statistic, we must instead refer to the sphericity-adjusted ϵ values.

We use the Greenhouse-Gaiser-adjusted ϵ because it is more conservative, and less likely to result in a type I error.

The GG-adjusted ϵ was 0.70, which was statistically significant, $p < .0001$, indicating that at least two time point had a different mean DEW score.

Post-hoc tests (pairwise t-tests adjusted with Bonferroni corrections) indicated that most mean differences were statistically significant, except for the differences between 10 years and 15 years of marriage, and those who are just married and married for 5 years.

Most mean differences, although significant, were rather small: given the wide range of WES scores, the largest mean difference was only 3.87 units, which was very small, and practically nearly meaningless.

Thus, men on average tend to very slightly open up to their wives regarding their desires to express worries over time, though such a difference may not be large enough to matter.